

Introduction to Programming

COMP101

2025-26

Lab-06

Supervised

Objectives: Nested loops

Exercise-01: for in for

Exercise-02: while in while

Exercise-03: if in while

Exercise-01: Debug

In the following code, all the code has been left-aligned due to poor syntax.

Debug the following so it produces output.

```
num_of_tables = int(input("How many tables do you want to print? "))
for i in range (1,num_of_tables+1)
for j in range (1,11)
print(i,"*",j,"=",i*j)
print()
```

Exercise-01 SOLUTION:

```
num_of_tables = int(input("How many tables do you want to print? "))
for i in range (1,num_of_tables+1):
    for j in range (1,11):
        print(i,"*",j,"=",i*j)
    print()
```

Try replacing the range statements with some different numbers to see what it does to the output.

Exercise-02: Debug

The following code has indentation issues.
Debug to give correct execution.

```
student = input('Students y/n: ')
while student == 'y':
    work = input ('Work to grade y/n: ')
    while work == 'y':
        grade = int(input('Enter grade: '))
        work = input ('More work y/n: ')
student = input ('Another student y/n: ')
```

Exercise-02: SOLUTION

```
student = input('Students y/n: ')
while student == 'y':
    work = input ('Work to grade y/n: ')
    while work == 'y':
        grade = int(input('Enter grade: '))
        work = input ('More work y/n: ')
student = input ('Another student y/n: ')
```

Exercise-03: Debug

The following code has indentation issues.
Debug to give correct execution.

```
student = input("Students y/n: ")
while student == 'y':
    grade = int(input('Enter grade: '))
    if grade >=40:
        print ('pass')
    else:
        print ('fail')
    student = input ('Another student y/n: ')
```

Exercise-03: SOLUTION

```
student = input("Students y/n: ")
while student == 'y':
    grade = int(input('Enter grade: '))
    if grade >=40:
        print ('pass')
    else:
        print ('fail')
    student = input ('Another student y/n: ')
```

Test the program with grades around the ‘grade boundary’ to check the code is not ‘off-by-one’, e.g. 41, 40, 39, -1, 0, 1